Environmental Engineering

Unit I: Water Supply

Syllabus

Estimation of surface and subsurface water resources - Predicting demand for water - Impurities of water and their significance - Physical, chemical and bacteriological analysis - Waterborne diseases - Standards for potable water - Intake of water: pumping and gravity schemes.

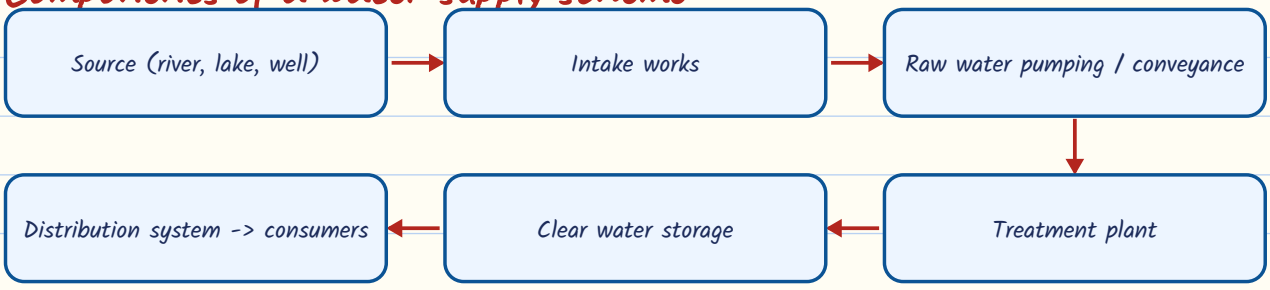
1. Introduction to Water Supply Engineering

Water supply engineering deals with collecting water from a source, treating it to make it safe, and delivering it to consumers in the **required quantity**, at **required pressure** and of **required quality**, at a reasonable cost.

Objectives of a water supply scheme

- To supply **safe and wholesome** (potable) water to the public
- To supply adequate quantity for domestic, industrial, public and fire-fighting needs
- To make water available at **all times** (continuous supply) and with sufficient pressure
- To keep the cost of construction, operation and maintenance **economical**
- To protect the public from waterborne diseases

Components of a water supply scheme



Types of water demand / needs

- **Domestic** - drinking, cooking, bathing, washing, flushing
- **Industrial** - cooling, boilers, process water
- **Public / civic** - road washing, parks, fountains, public toilets
- **Fire demand** - for fire fighting, needs high pressure and flow

- **Losses and wastes** - leakage, theft, meter errors (about 15%)

2. Sources of Water

All water on earth comes from rainfall (hydrologic cycle). Sources are classified into surface sources and subsurface (ground) sources.

Surface sources	Subsurface sources
Rivers, streams, lakes, ponds, impounding reservoirs	Wells, springs, infiltration galleries, infiltration wells
Water is **soft**, turbid, polluted, high bacteria	Water is **hard**, clear, low bacteria, more dissolved salts
Needs full treatment (sedimentation, filtration, disinfection)	Needs less treatment, often only disinfection
Large quantity available, variable with season	Quantity limited, depends on aquifer yield

Surface water sources

- **Rivers** - perennial rivers give large and continuous supply, but turbidity and pollution are high; intake works are needed
- **Lakes and ponds** - quiet water, suspended matter settles, but algae and weeds grow (taste and odour)
- **Impounding reservoirs** - a dam is built across a valley to store monsoon flow; used where rivers are non-perennial
- **Canals** - water is taken from irrigation canals; needs treatment

Subsurface (ground water) sources

- **Aquifer** - a water-bearing geological formation that stores and yields water (sand, gravel, fractured rock)
- **Aquiclude** - formation that stores but does not transmit water (clay)
- **Aquifuge** - neither stores nor transmits (solid granite)
- **Aquitard** - poorly permeable layer that transmits water slowly
- **Water table** - upper surface of the saturated zone; unconfined aquifer is under atmospheric pressure
- **Confined (artesian) aquifer** - bounded by impermeable layers, water is under pressure; artesian wells flow without pumping
- **Open (dug) wells** - shallow, large diameter, 5-10 m deep, get water from water table
- **Tube wells** - small diameter bore, 30-300 m; strainer tubes lower down

- **Springs** - natural outflow of ground water at surface
- **Infiltration galleries** - horizontal perforated pipes laid in water-bearing strata near rivers
- **Infiltration wells** - vertical wells sunk in the river bed or bank

3. Estimation of Surface Water Resources

The yield of a surface source depends on **rainfall**, catchment area and losses. For design of an impounding reservoir the **safe yield** is the maximum quantity available during the **driest period** (critical period).

Catchment area yield methods

- **Run-off** = Rainfall - (Evaporation + Infiltration + Transpiration + Depression storage)
- **Strange's table / curves** - run-off % based on catchment type (good, average, bad) and monsoon rainfall
- **Binnie's percentages** - run-off in terms of rainfall (based on Nagpur data)
- **Barlow's tables** - run-off coefficient (k) for different catchment classes: $Q = k \times P$
- **Inglis and De Souza formula** (for Western Ghats / Deccan): $R = 0.85\,P - 30.5$ (R, P in cm)

Rational formula (for small catchments)

$Q = C\,i\,A / 360$ (Q in m^3/s, i in mm/h, A in ha)

C = run-off coefficient (0.1 for forest to 0.9 for paved areas)

Storage capacity of reservoir (mass curve method)

- Plot cumulative inflow against time (**mass curve**)
- Draw tangents with slope equal to the demand rate (demand lines) from the peaks of the mass curve
- The maximum vertical intercept between the demand line and the mass curve gives the **required storage capacity**
- Reservoir capacity must also allow for evaporation, silting, and dead storage

4. Estimation of Subsurface (Ground) Water Resources

Darcy's law

Darcy's Law

$v = K\ i$ and $Q = K\ i\ A = K\ (dh/dL)\ A$

K = coefficient of permeability (m/day), i = hydraulic gradient, A = area of flow

Valid for laminar flow, Reynolds number less than 1 (up to about 10)

Specific yield and specific retention

- **Porosity (n)** = volume of voids / total volume
- **Specific yield (S_y)** = volume of water that drains by gravity / total volume of soil
- **Specific retention (S_r)** = water retained against gravity; $n = S_y + S_r$
- **Storage coefficient** = volume of water released per unit area per unit fall in head

Yield of wells

Steady flow into wells (Dupuit's theory)

Unconfined aquifer

$Q = pi\ K\ (h_2^2 - h_1^2)\ /\ ln(r_2/r_1)$

h_1, h_2 = water heights in observation wells at radial distances r_1, r_2

Confined aquifer

$Q = 2\ pi\ K\ b\ (h_2 - h_1)\ /\ ln(r_2/r_1)$

b = thickness of the confined aquifer

- **Specific capacity** of a well = Q / s (yield per unit drawdown)
- **Radius of influence** - distance from the well to where drawdown is zero
- **Cone of depression** - funnel shaped drop in water table around a pumped well
- Safe yield of a ground water basin = recharge - natural discharge

5. Population Forecasting and Water Demand

A water supply scheme is designed for a **design period** of 20-30 years (intake, treatment 15 years; pipes 30 years; pumps 10-15 years). So the future population must be predicted.

Methods of population forecasting

Method	Formula / idea	Used for
Arithmetic increase	$P_n = P + n \times$ (average increment per decade)	Old, large, stable cities
Geometric increase	$P_n = P (1 + r/100)^n$	Growing young cities
Incremental increase	$P_n = P + n \times X + n(n+1)/2 \times Y$ (X = avg increase, Y = avg of increments)	Average cities
Decreasing rate of growth	Rate of growth decreases with time, saturation limit assumed	Cities nearing saturation
Graphical method	Curve of past data extended / comparison with similar cities	Quick check
Logistic curve	$P = P_s / (1 + m\, e^{nx})$	Long term, S-shaped growth
Ratio / apportionment	Ratio of city to state population kept constant	Where state data known

Per capita demand (as per IS 1172 / CPHEEO)

Type of town	Water demand (lpcd)
Towns with piped supply, no sewerage	70
Towns with piped supply and sewerage	135
Metropolitan and mega cities	150 (CPHEEO manual up to 150-200)

Typical break-up of domestic demand (135 lpcd)

* Drinking 5, Cooking 5, Bathing 55, Washing clothes 20
* Washing house 10, Flushing of WC 30, Others 10
* Total 135 lpcd (plus 15% losses)

Fire demand formulae (Q in kilolitres/day, P in thousands)

- **Kuichling:** $Q = 3182 \sqrt{P}$
- **Freeman:** $Q = 1136 (P/5 + 10)$
- **National Board of Fire Underwriters:** $Q = 4637 \sqrt{P} (1 - 0.01 \sqrt{P})$
- **Buston:** $Q = 5663 \sqrt{P}$

Variations in demand

- **Maximum daily demand** = 1.8 x average daily demand
- **Peak hourly demand** = 1.5 x maximum daily demand = 2.7 x average
- Seasonal (summer high), daily (weekends), hourly (morning and evening peaks)
- Distribution system is designed for peak hourly demand; pumps for maximum daily demand

Factors affecting per capita demand

- Size of city, climate, standard of living, industrial and commercial activity
- Quality of water, pressure in mains, cost of water (metered supply cuts demand)
- Systems of supply (continuous or intermittent), sewerage facility, habits of people

Example (arithmetic method)

Populations: 1991 = 40,000; 2001 = 45,000; 2011 = 52,000. Find the 2031 population.

Average increment = (5000 + 7000)/2 = 6000 per decade

P_{2031} = 52,000 + 2 x 6000 = **64,000**

6. Impurities of Water and Their Significance

Natural water is never pure. It picks up impurities from air, soil and human activity. Impurities are classified as **suspended, colloidal** and **dissolved** (by particle size, >1 micron, 1 nm - 1 micron, <1 nm).

Impurity	Source	Significance
Suspended solids (silt, clay, algae)	Erosion, run-off	Turbidity, shields bacteria, removed by sedimentation and filtration
Colloids	Organic decay, clay	Colour and turbidity, need coagulation
Dissolved gases (O_2, CO_2, H_2S)	Air, decay	Corrosion, taste and odour
Calcium, magnesium	Rocks (limestone, dolomite)	Hardness, scale in boilers, soap wasted
Iron, manganese	Ground water, minerals	Red/black stains, bitter taste, bacterial growth
Chlorides	Sea water, sewage	Salty taste, indicates pollution
Sulphates	Gypsum, industrial waste	Laxative effect (above 400 mg/L), corrosion of concrete
Nitrates	Fertilizers, sewage	**Methaemoglobinemia (blue baby)** in infants
Fluoride	Fluorite-bearing rocks	Below 0.6 mg/L dental caries; above 1.5 mg/L **fluorosis**
Arsenic, lead, mercury, cadmium	Industry, geological	Toxic, carcinogenic, nervous system damage
Pathogens	Human and animal excreta	Waterborne diseases

7. Analysis of Water

A. Physical characteristics

- **Turbidity** - cloudiness due to suspended matter; measured by **nephelometer** in NTU; limit 1 NTU (acceptable), 5 NTU (permissible)
- **Colour** - true colour (dissolved) and apparent colour; measured by Tintometer/platinum-cobalt scale, limit 5 Hazen units (15 permissible)
- **Taste and odour** - from algae, decaying matter, chemicals; threshold odour number test
- **Temperature** - best 10-15 degrees C; affects chemical reactions, bacteria and dissolved oxygen
- **Electrical conductivity** - gives idea of total dissolved solids; **TDS** limit 500 mg/L (2000 permissible)
- **pH** - measured by pH meter; limit 6.5 to 8.5

B. Chemical characteristics

- **Alkalinity** - capacity to neutralize acids (HCO_3, CO_3, OH); titration with H_2SO_4 using phenolphthalein and methyl orange
- **Acidity** - capacity to neutralize bases (free CO_2, mineral acids)
- **Hardness** - due to Ca and Mg; **temporary** (carbonate, removed by boiling) and **permanent** (sulphate, chloride); EDTA titration with Eriochrome Black-T indicator; limit 200 mg/L (600 permissible)
- **Chlorides** - Mohr's method (silver nitrate titration, potassium chromate indicator); limit 250 mg/L
- **Dissolved oxygen (DO)** - Winkler's iodometric method
- **BOD** - oxygen used by bacteria to stabilize organic matter in 5 days at 20 degrees C
- **COD** - oxygen needed for chemical oxidation (dichromate reflux), COD > BOD
- **Residual chlorine** - orthotolidine (OT) test, DPD method
- **Nitrogen forms** - free ammonia (fresh pollution), albuminoid, nitrite (partly decomposed), nitrate (fully oxidised, old pollution)
- **Metals** - Fe, Mn, Pb, As by atomic absorption spectrophotometer (AAS)

Hardness classification

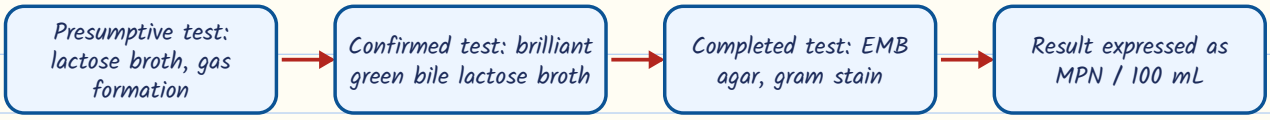
* 0 - 75 mg/L as $CaCO_3$: soft

* 75 - 150 : moderately hard
* 150 - 300 : hard
* above 300 : very hard

C. Bacteriological characteristics

Testing for every pathogen is difficult, so we test for **indicator organisms** - the coliform group (**E. coli**) which live in the human intestine and are present in large numbers in faeces. Presence of coliforms indicates faecal pollution.

- **Standard Plate Count** - total bacteria colonies on nutrient agar at 37 degrees C for 24 h, should be less than 500 per mL
- **MPN test (Most Probable Number)** - multiple tube fermentation: presumptive, confirmed and completed tests
- **Membrane filter technique** - sample filtered through 0.45 micron membrane, colonies counted on m-Endo media
- **Colilert / H_2S strip test** - quick field tests for presence/absence
- Drinking water standard: **zero coliform / E. coli per 100 mL**



8. Waterborne Diseases

Type	Disease	Causative agent
Bacterial	Cholera	Vibrio cholerae
Bacterial	Typhoid	Salmonella typhi
Bacterial	Bacillary dysentery	Shigella
Bacterial	Gastroenteritis	E. coli
Viral	Infectious hepatitis (Jaundice)	Hepatitis A / E virus
Viral	Poliomyelitis	Polio virus
Protozoal	Amoebic dysentery	Entamoeba histolytica
Protozoal	Giardiasis	Giardia lamblia
Helminthic	Guinea worm, round worm, tape worm	Dracunculus, Ascaris, Taenia
Water-based / vector	Malaria, dengue, filariasis (mosquitoes breed in water)	Parasites / viruses

- Spread by **faecal-oral route**: drinking contaminated water, food, flies
- Prevention: protected water source, treatment and disinfection, sanitary disposal of excreta, personal hygiene, chlorination, boiling of water

9. Standards for Potable Water

Potable water should be colourless, odourless, tasteless, free from pathogens, and have no harmful chemicals. Indian standards: **IS 10500:2012** (BIS) and the CPHEEO manual; also WHO guidelines.

Parameter	Acceptable	Permissible (no alternate source)
pH	6.5 – 8.5	No relaxation
Turbidity	1 NTU	5 NTU
Colour	5 Hazen	15 Hazen
TDS	500 mg/L	2000 mg/L
Total hardness	200 mg/L	600 mg/L
Chloride	250 mg/L	1000 mg/L
Sulphate	200 mg/L	400 mg/L
Fluoride	1.0 mg/L	1.5 mg/L
Iron	0.3 mg/L	No relaxation
Nitrate	45 mg/L	No relaxation
Arsenic	0.01 mg/L	0.05 mg/L
Residual chlorine	0.2 mg/L	1.0 mg/L
Coliform / E. coli	Absent in 100 mL	Absent

10. Intake of Water

An **intake** is a structure placed in the source to withdraw water and carry it by pipe or conduit to the treatment plant. A good intake gives water of the **best quality** at **all seasons**.

Factors for selecting the site of intake

- Should be **upstream** of the town and sewage outfalls, to get purest water
- Water available throughout the year, even in the driest period
- Located where the **river bed is stable** (not shifting), on the concave bank (outer side of a bend) where deep water is present

- Easy access in floods, safe against flood and ice, away from navigation channel
- Near the treatment plant to cut the cost of conveyance
- Free from floating matter, silt, and weeds

Types of intakes

Intake	Description	Suited to
Submerged intake	Pipe or chamber fully under water	Rivers and small lakes, economical
Wet intake (tower)	Masonry tower with openings (ports) at different levels, water enters the tower and is drawn by pump from a sump	Large rivers, deep lakes, reservoirs
Dry intake (tower)	Tower dry inside, water flows through ports to a conduit, valves inside tower	Deep, large lakes
Canal intake	Opening taken from a canal	Where canal water is used
Reservoir / dam intake	Draw-off at different levels through the dam body (sluice)	Impounding reservoirs
Floating / Jack well	Intake on a raft or well on bank with inlet pipes	Variable river level

- **River intake** - an intake well is built in bank or river, connected by pipes, with a **bell-mouth** and screen to prevent entry of fish and debris
- **Intake ports** are provided at various levels so water can be taken from the layer of best quality
- **Screens (coarse)** - bar screens to exclude floating matter; velocity through screen not more than 0.15 m/s
- Valves are provided to regulate and cut off flow

11. Conveyance of Raw Water: Gravity and Pumping Schemes

Water from intake is moved to the treatment plant, from there to service reservoirs, by one of the following systems.

Gravity system	Pumping system
Source is **higher** than the town, water flows by gravity	Source is **lower** than the town, pumps are used
Most economical and reliable, no power cost	Needs power, repair and maintenance, standby pumps

Gravity system	Pumping system
Needs pipes capable of carrying pressure; velocity 0.6 – 1.5 m/s	Pump may be directly to the distribution system or to elevated storage
Suitable for hill and spring sources	Typical for river and tube-well sources
Can be **gravity main** or **gravity + pumping combined**	Combined: pumping to a reservoir then gravity distribution is best

Pumping schemes

- **Direct pumping** – pumps feed the distribution system directly; no storage; pressure fluctuates; standby units needed; used for small towns
- **Pumping with elevated storage** – pumps work at uniform rate, excess goes to the service reservoir, and the reservoir meets the peak demand; most favoured
- **Pump types:** centrifugal pumps (most common), reciprocating, submersible (tube wells), air-lift pumps
- **Pump head** = static lift + friction losses + minor losses + residual head
- Power (kW) = (Q x H x gamma) / (eta x 1000) = 9.81 Q H / eta (Q in m^3/s, H in m, eta = overall efficiency)

Pipes for conveyance

- **Cast iron, ductile iron** – strong, long life (100 yrs), but heavy
- **Steel** – high pressure, light, welded; needs protection from corrosion
- **PVC / HDPE** – light, corrosion free, smooth, cheap, used for small diameters
- **RCC / prestressed concrete** – large diameter gravity mains
- **Asbestos cement** – now avoided because of health risk

Economical diameter of rising main

Lea's formula

$D = 1.22 \sqrt{Q}$ (D in m, Q in m^3/s)

Lamont / Marcel formula: D = 0.97 to 1.5 $\sqrt{Q}$, based on balancing pipe cost and pumping cost

12. Quick Revision Points

Remember for exam

* Design period: intake 30 yrs, pipes 30 yrs, pumps 15 yrs, treatment units 15 yrs
* Domestic demand 135 lpcd; max daily = 1.8 x avg; peak = 1.5 x max daily
* Geometric method for growing cities, arithmetic for old and saturated cities
* BOD uses 5 days at 20 degrees C; coliform standard is zero per 100 mL
* Fluoride limit 1.0 - 1.5 mg/L; nitrate 45 mg/L; TDS 500 mg/L; hardness 200 mg/L
* IS 10500:2012 is the Indian drinking water standard
* Intake is placed upstream of the town on the concave bank

Important questions

1. Explain the various sources of water supply with their merits and demerits.
2. Discuss the methods of forecasting population. Which is suitable for which city?
3. Explain the factors affecting per capita demand and the variations in demand.
4. Describe the physical, chemical and bacteriological tests for water quality.
5. What are waterborne diseases? Explain the preventive measures.
6. Explain IS 10500 drinking water standards for important parameters.
7. Describe the types of intake structures with neat sketches and the factors for selecting the site.
8. Compare gravity and pumping schemes of conveyance of water.
9. A city has population 25,000 (1991), 30,000 (2001), 38,000 (2011). Estimate the population in 2031 by the geometric and incremental increase methods.